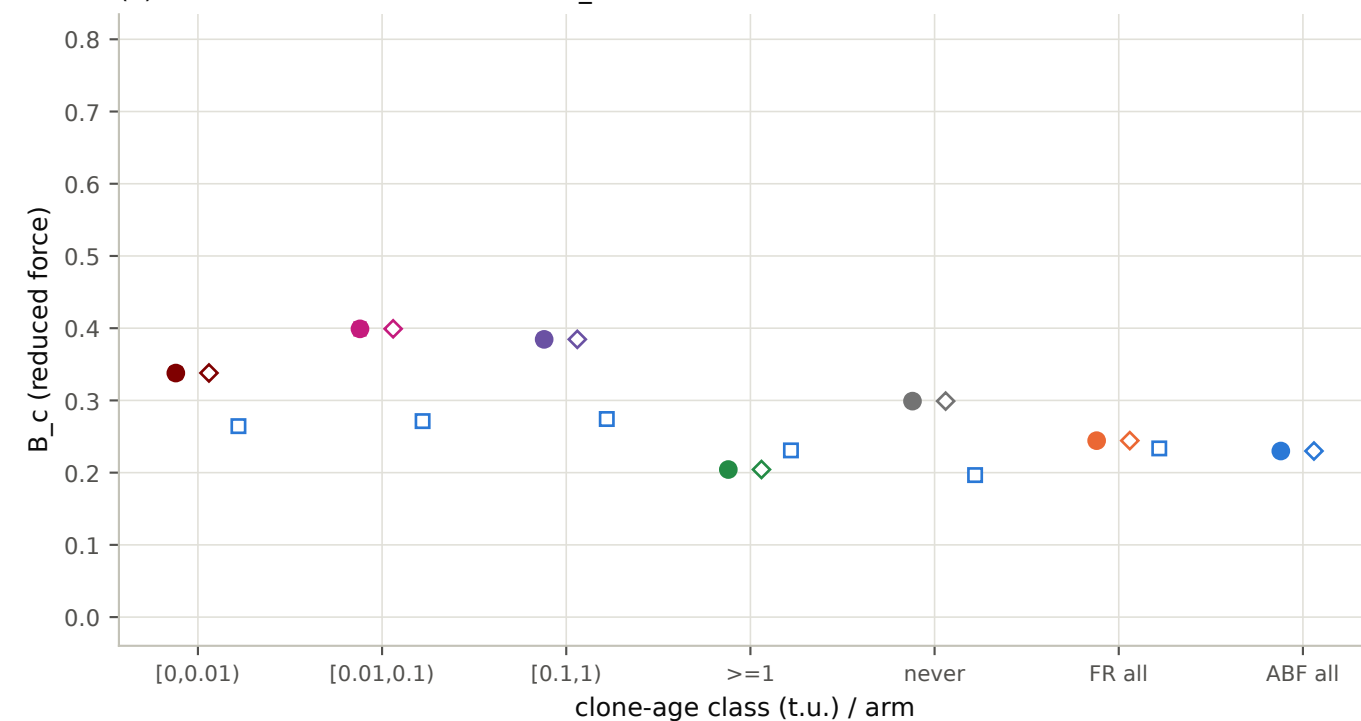
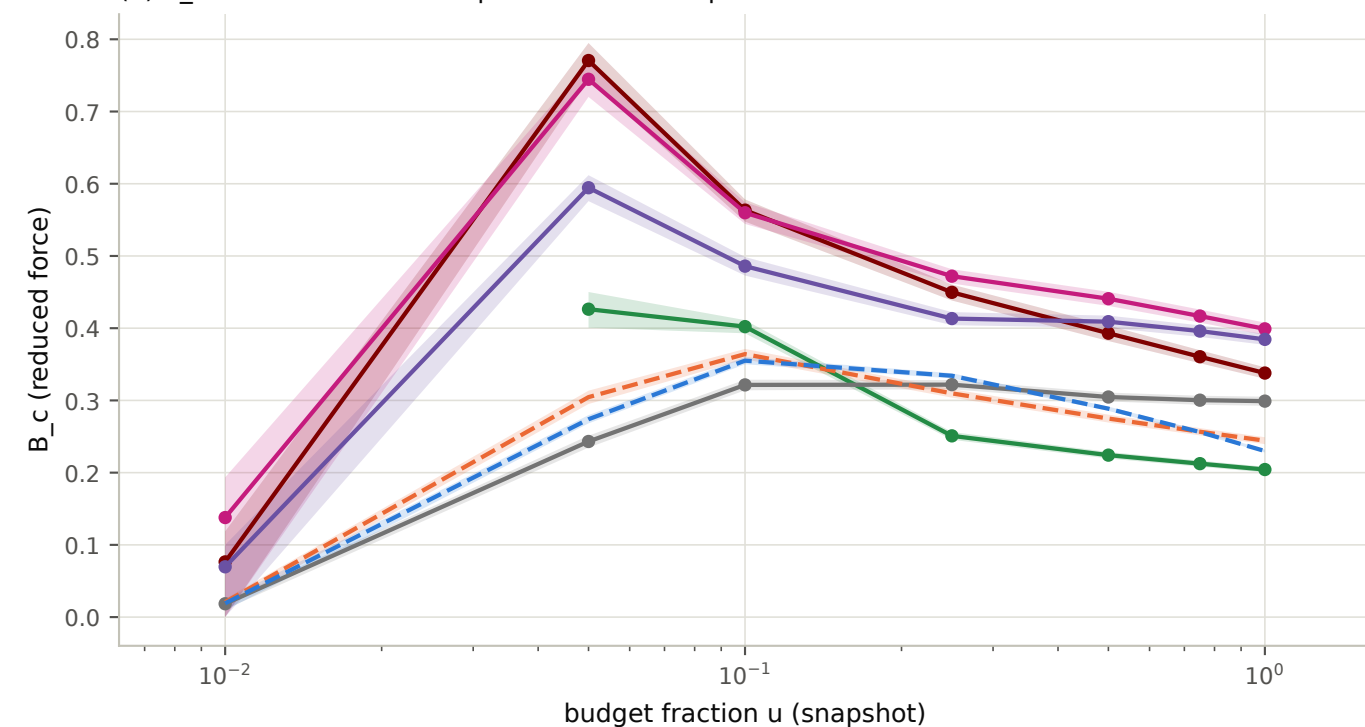


D2(ii) class-wise conditional mean-force error vs the bin-averaged  $F^*$ : l: shifted fibre,  $\kappa = 1$  (variance-matched control; constant width, curved slow fibre),  $N = 2048$   
 $B_c^2 = \sum_j n_{cj} [(m_{cj} - \langle F^* \rangle_j)^2 - V_{cj}] / \sum_j n_{cj}$  over eval bins ( $V_{cj}$ : seed-cluster variance of the pooled bin mean); CIs: 10000 seed-bootstrap resamples (ABF+FR  $n = 32$ , ABF  $n = 32$ ;  $B_c$ : basic interval of the debiased  $B_c^2$ ). The ABF arm is the reference level. y ranges shared by every cell of this experiment at this N.

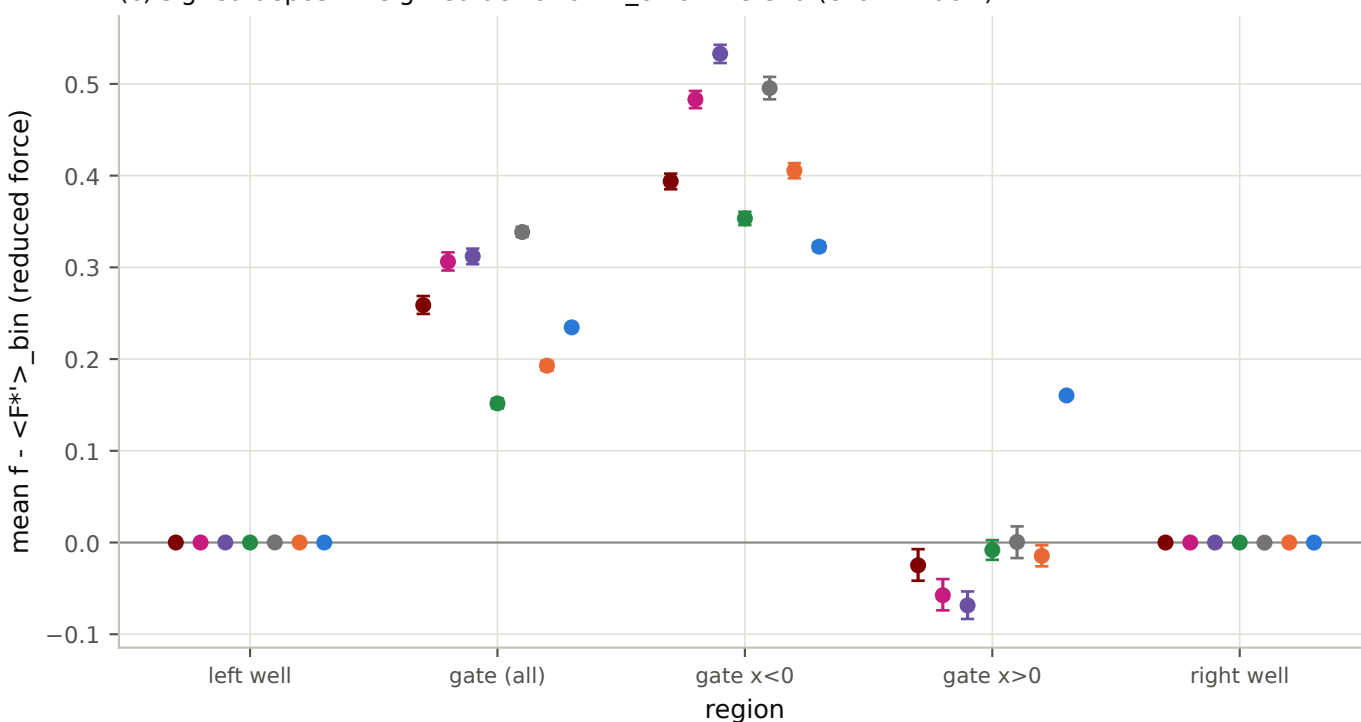
(a) debiased RMS mean-force bias  $B_c$  at the end



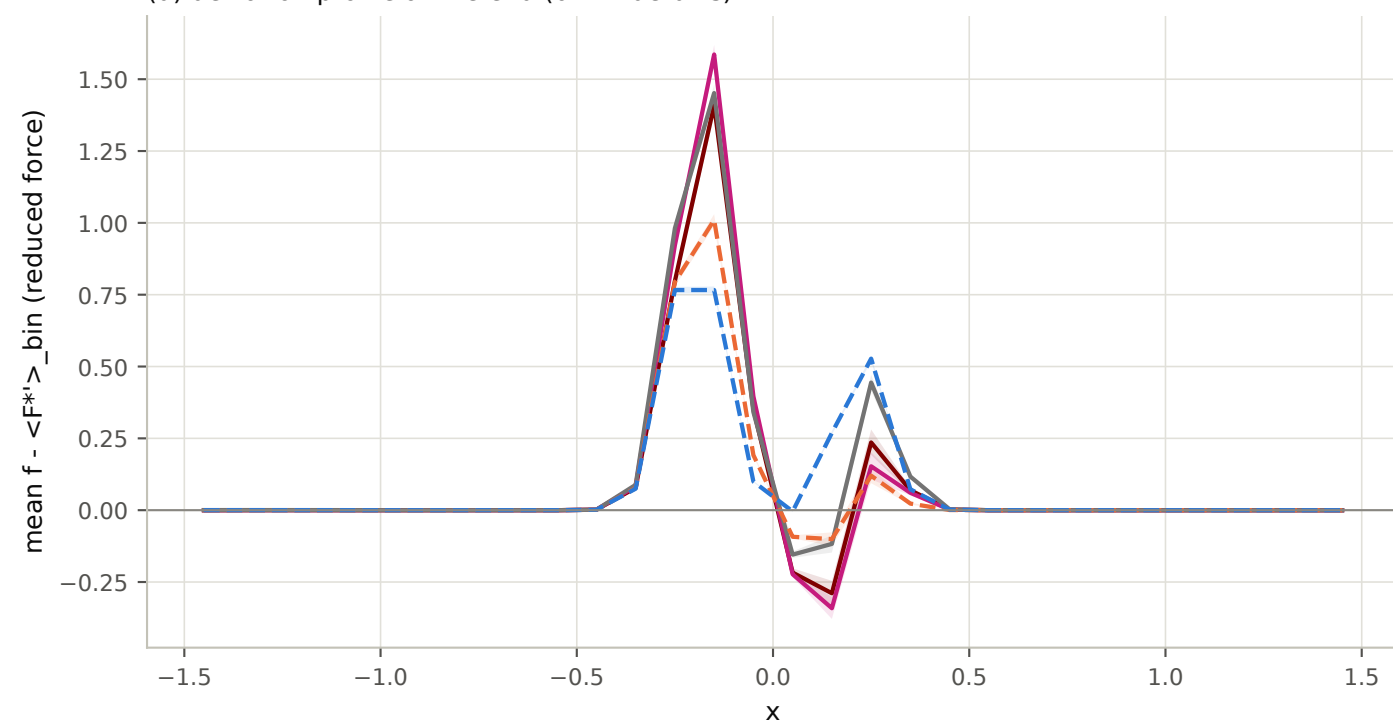
(b)  $B_c$  of the cumulative deposits at each snapshot



(c) signed deposit-weighted deviation  $D_{cR}$  at the end (eval window)



(d) deviation profile at the end (0.1-wide bins)



— clone age [0,0.01) t.u.    — clone age [0.1,1) t.u.    — never cloned    - - ABF (all deposits = never cloned)    □ (a) EXPLORATORY: ABF's B at the class's positions (matched, paired)  
— clone age [0.01,0.1) t.u.    — clone age  $\geq 1$  t.u.    - - ABF+FR, all deposits    ◇ raw RMS (not debiased)     95 % seed-bootstrap CI